RAG Knowledge Assistant: Requirements Specification

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Executive Summary

This document specifies the requirements for an institutional knowledge assistant built on Retrieval-Augmented Generation (RAG). Staff and students will ask questions in natural language and receive answers grounded in documents approved by the institution, with clear source references. A single administrator controls which PDF documents enter the knowledge base, ensuring that answers draw only on trusted material.

The system is a web application with a Python FastAPI backend and a browser-based frontend. A sidebar offers two views: **Admin Upload** and **Chat**. Retrieval combines semantic and keyword search, with optional reranking, to deliver accurate and relevant answers.

1. Project Overview and Goals

Purpose: Provide a reliable, self-service way to find answers in institutional documents such as policies, handbooks, regulations, course guides and research material.

Goals

- Reduce the time users spend searching long documents manually.
- Provide answers that are grounded in approved content and traceable to a source.
- Give administrators simple, exclusive control over the knowledge base.
- Offer an intuitive chat experience that needs no training.
- Reduce the volume of repetitive questions sent to staff.

Success measures: High answer relevance, low rate of unsupported answers, and positive user feedback during pilot (see Section 9).

2. Key Users and Roles

Role	Description	Key permissions
Administrator	Designated knowledge manager	Upload and manage PDFs; access Admin Upload view; use Chat
End User	Student, faculty or staff member	Use Chat; view own conversation history
System Owner / IT	Hosts and maintains the system	Deployment, monitoring, backups (outside application UI)

3. Functional Requirements

3.1 Document Management (Admin only)

- **FR-1:** The system shall allow only the Administrator to upload PDF documents.
- **FR-2:** The system shall validate each upload (file type, size limit, readable text) and report success or failure clearly.
- **FR-3:** The system shall show the Administrator a list of uploaded documents with name, upload date and processing status.
- **FR-4:** The system shall allow the Administrator to remove a document and all of its associated content from the knowledge base.

3.2 Retrieval

- **FR-5:** The system shall store processed document content in a vector database to enable semantic search.
- **FR-6:** The system shall support hybrid retrieval, combining semantic similarity with keyword-based matching to find the most relevant document passages.
- **FR-7:** The system shall optionally support a reranking step that reorders candidate passages by relevance before answer generation.
- **FR-8:** The system shall retrieve a limited set of top-ranked passages for each question.

3.3 Answer Generation

- **FR-9:** The system shall generate answers using only the retrieved passages as the basis for its response.
- **FR-10:** The system shall cite the source document (and page where available) for each answer.
- **FR-11:** When retrieved content is insufficient, the system shall state that it cannot find a reliable answer rather than guess.

3.4 Chat Interface and History

- **FR-12:** The system shall provide a chat view where users type questions and receive responses in a conversational layout.
- **FR-13:** The system shall keep a simple conversation history within a session so follow-up questions retain context.

- **FR-14:** Users shall be able to start a new conversation and clear the current one.

3.5 Navigation

- **FR-15:** The application shall include a sidebar with two views: **Admin Upload** and **Chat**.
- **FR-16:** The Admin Upload view shall be visible and accessible only to the Administrator.

4. Non-Functional Requirements

- **Usability:** Clean, consistent interface; a first-time user can ask a question without instructions.
- **Reliability:** Failed document processing must not affect existing content or chat availability.
- **Maintainability:** Modular components (ingestion, retrieval, generation, interface) so each can be updated independently.
- **Compatibility:** Works on current versions of major desktop and mobile browsers.
- **Scalability:** Designed to grow from a pilot collection to a larger institutional library without redesign.
- **Transparency:** Answers always show their sources so users can verify them.
- **Accessibility:** Follows common accessibility practices (keyboard navigation, readable contrast).

5. System Architecture Summary

The solution follows a straightforward layered design:

- **Frontend UI:** Web interface with a sidebar, an Admin Upload view and a Chat view.
- **FastAPI backend:** Exposes services for authentication, document upload, chat and history.
- **Ingestion pipeline:** Extracts text from PDFs, divides it into chunks and creates embeddings.
- **Vector database:** Stores embeddings and chunk metadata for semantic search, alongside a keyword index.
- **Retrieval layer:** Merges semantic and keyword results, with optional reranking.
- **Language model service:** Produces the grounded answer from the retrieved passages and the user's question.
- **Session store:** Holds conversation history.

Typical flow: User question > hybrid retrieval > optional reranking > grounded answer with citations > displayed in chat.

6. Data Handling and Document Processing

- **Accepted input:** Text-based PDFs up to an agreed size limit.
- **Processing steps:** Text extraction, cleaning, splitting into overlapping chunks, embedding, and indexing for both semantic and keyword search.
- **Metadata retained:** Document title, upload date, page reference and chunk identifier to support citations.
- **Status tracking:** Each document shows as Processing, Ready or Failed.
- **Deletion:** Removing a document removes its chunks and index entries so it no longer appears in answers.
- **Conversation data:** Stored only as needed for session context and retained for an agreed period.

7. Security and Access Control

- **Authentication:** Users must sign in before using the system; method (for example institutional single sign-on) to be confirmed.
- **Role-based access:** Only the Administrator role can upload or delete documents; this is enforced on the server, not just hidden in the interface.
- **Data protection:** Encrypted connections (HTTPS) and protected storage of documents, embeddings and credentials.
- **Input safeguards:** Upload validation, file scanning where available, and protection against malicious files and prompt manipulation.
- **Audit logging:** Uploads, deletions and sign-in events are recorded with user and time.
- **Privacy:** Users cannot view other users' conversations.

8. Performance Expectations

Area	Target (to be confirmed during pilot)
Chat response (first output)	Within 5 seconds for typical questions
Complete answer	Within 15 seconds under normal load
Document processing	A 100-page PDF ready in under 5 minutes
Concurrent users	At least 25 simultaneous chat users in the pilot
Availability	99% during institutional working hours

9. Acceptance Criteria

- Only the Administrator can upload or delete documents; other users are denied access.
- Uploaded PDFs become searchable and are reflected in answers after processing completes.
- At least 85% of a reviewed set of test questions receive relevant, source-supported answers.
- Every answer displays at least one source reference.
- Questions outside the knowledge base receive a clear "cannot find an answer" response.
- Follow-up questions correctly use earlier messages in the same conversation.
- Hybrid retrieval returns relevant results for both conceptual questions and exact terms (such as codes or names).
- The sidebar shows the correct views for each role.
- Performance targets in Section 8 are met in pilot testing.

10. Scope Limitations and Assumptions

In scope: PDF ingestion by one administrator, hybrid retrieval, grounded chat answers with citations, session-level conversation history, and role-based access.

Out of scope (initial release)

- File formats other than PDF (Word, web pages, images or scanned documents requiring OCR).
- Multiple administrator tiers, approval workflows or per-document permissions.
- Long-term personalization, analytics dashboards and cross-device history sync.
- Mobile native apps, voice input and multilingual translation.
- Automatic updating of documents from external systems.

11. Assumptions and Risks

Assumptions

- Source documents are authoritative, current and contain extractable text.
- The institution will provide hosting, access to a language model service and user identity management.
- A single Administrator is sufficient for the initial release.

Risks and mitigations

- **Inaccurate or unsupported answers:** Restrict answers to retrieved content, require citations, and allow "no answer" responses.
- **Poor-quality or scanned PDFs:** Validate uploads and flag unreadable files; consider OCR in a later phase.
- **Outdated documents:** Show upload dates and encourage periodic review by the Administrator.
- **Sensitive content exposure:** Enforce access control and limit uploads to approved material.
- **Response delays under load:** Use configurable retrieval limits and optional reranking only where it adds value.
- **Service cost or availability of the language model:** Monitor usage and define a fallback plan.

Final Acceptance Checklist

- ■ Administrator-only upload is enforced and verified
- ■ PDFs are processed, stored and searchable in the vector database
- ■ Hybrid retrieval (semantic and keyword) works as specified
- ■ Reranking is documented as optional and configurable
- ■ Answers are grounded and include source citations
- ■ Insufficient-information cases return a clear message
- ■ Chat interface and simple conversation history function correctly
- ■ Sidebar presents Admin Upload and Chat views by role
- ■ Security, logging and privacy controls are in place
- ■ Performance targets are met in pilot testing
- ■ Stakeholder sign-off obtained

Approved by: _____ **Date:** _____